

CONFIDENCE IS NOT COMPETENCE: INVERSE-FOLDING MODELS KNOW BINDING IN THE MIXED DERIVATIVE

Anonymous authors

Paper under double-blind review

ABSTRACT

Staged binder design generates a backbone and then inverse-folds a sequence, and the field reads an inverse-folding model’s *confidence* to judge its interface choices. We show this reads binding from the wrong place. Confidence and binding competence are two derivatives of one likelihood: confidence is the *diagonal*, any scalar of the bound-conditioned distribution P , while binding *leverage* is the *mixed second derivative*, the response to ablating the partner. We prove leverage is not a function of P , so every scalar the field reads (recovery, confidence, a complex-versus-monomer KL) is blind to binding **by construction**, and a flexible learner over all of P recovers only 37% of it. The consequences are concrete and controlled. On SKEMPI natural complexes, confidence ranks hotspots at chance, whereas the mixed derivative, a zero-shot readout that uses no binding labels, adds binding signal beyond geometry, evolutionary conservation, and the one-pass log-odds (Spearman -0.30 with experimental $\Delta\Delta G$, conditional predictive impact $+0.059$) across four inverse-folding architectures, and beyond a fitted physics energy function (partial -0.17). The direction is actionable: an $+\alpha \cdot L$ tilt on a frozen ProteinMPNN raises binding-favorability against a matched random control, confirmed by an independent sequence model, two structure predictors, and FoldX, on crystal and on predicted backbones. The published hotspot deficit is largely a burial confound. The mixed derivative is the model’s **classifier-free-guidance direction**, a general recipe: to read an untrained quantity off a conditional generative model, ablate the conditioner. All code and analysis-ready data reproduce every number and figure.

1 INTRODUCTION

The dominant paradigm for designing protein binders is staged: a generative model proposes a backbone, and an inverse-folding model assigns a sequence by maximising $p(\text{sequence} \mid \text{backbone})$. This factorisation is convenient, but it omits the very quantity a binder exists to optimise. There is no binding-energy term anywhere in $p(\text{sequence} \mid \text{backbone})$. The concern this raises is sharpest at *interface hotspots*: the small set of residues that contribute most of the binding free energy, and which are frequently *frustrated*, buried polar residues or strained rotamers that are locally unfavourable for the monomer’s fold but bought because they pay off in binding. If inverse folding optimises fold-compatibility, the reasoning goes, it should systematically miss exactly these residues, and a hotspot should sit in the tail of the model’s distribution rather than at its mode. A prominent measurement appears to confirm this: ProBID-Net (Chen et al., 2024) reports inverse-folding recovery of 0.334 at hotspots versus 0.472 elsewhere, and attributes the gap to dynamics.

This paper asks one question: when an inverse-folding model is used to build a binder, where, if anywhere, does its knowledge of *binding* live, and can a designer read it and act on it? The scope is deliberately narrow and practical: structure-conditioned inverse-folding models (ProteinMPNN, ESM-IF1, and three others), protein-protein interface binding, and readouts computable at design time. Our answer is that the binding signal is real, but the field has been reading it from the wrong place. It has read *scalar* summaries of the model’s output (recovery, confidence, a complex-vs-monomer KL) that are blind to binding **by construction**, when it lives in a **mixed second derivative** that is cheap to compute and directly actionable.

The consequence is immediate for practice, in three forms. As a **diagnostic**: inverse-folding confidence is a geometry detector, not a binding score, so a pipeline that distrusts it at the interface (as the field’s leading one-shot designer does, by freezing the interface) is right to, and can now do better. As a **training-free ranker**: the mixed derivative locates interface hotspots and adds on top of the exact feature set published predictors already use, at no training cost. As a **drop-in steering knob**: a $+\alpha \cdot L$ tilt biases a *frozen, off-the-shelf* inverse-folding model toward higher-binding interface residues with no retraining, confirmed by an independent sequence model, an independent structure predictor, and a physics energy function. The recipe is not protein-specific. The mixed derivative is the model’s **classifier-free-guidance direction**, so the same move (ablate the conditioner, read the mixed derivative, not the marginal) applies to any conditional generative model, which is what places the result at a representation-learning venue rather than a purely methodological one.

We are precise about what is new, because the nearest prior methods already use this operator. The leverage operator itself, a partner-ablated second forward pass, is the BA-Cycle operator of Jiao et al. (2025), and decoders such as RedNet (Xie et al., 2026) already tilt along this direction. Our contribution is not the operator but what it means and what it buys. We give a decomposition that separates confidence from leverage at the mechanism level, a proof that no scalar of the bound distribution can recover leverage, the first control showing the signal survives beyond geometry and beyond evolutionary conservation (the full feature set published hotspot predictors use), a feature-class law that holds across four inverse-folding architectures, and steering of a frozen model that three independent instruments confirm. Where prior work uses the direction, we characterise it, bound what it can and cannot express, and show that a designer can read it and act on it.

A single principle runs through all of this: confidence is not competence, and it recurs at every level of the modeling stack. An inverse-folding model’s confidence (any scalar of its bound distribution) is blind to binding by construction. A structure predictor’s interface ipTM, confidence one level up, is blind to binding-*specificity*. It rewards a confidently-packed interface, which a mere foldability tilt achieves, so a be-more-confident tilt beats the binding direction on ipTM while losing to it under models with an explicit binding term. It takes a readout with such a term, the mixed derivative or a physics energy function, to see binding at all. The competence always lives in the response to ablating the partner, never in the bound-state confidence, whichever model does the reading.

There is a confound that no prior analysis controls, and it runs opposite to intuition. Hotspots are, on average, *more deeply buried* than other interface residues, and burial is precisely where inverse folding is *most* confident and accurate. An uncontrolled hotspot-vs-rest comparison therefore mixes a putative binding effect with a large burial effect of the opposite sign, so the naive comparison **hides** any real deficit rather than inventing one. Controlling burial is not a detail. It is the experiment.

Contributions.

- **A decomposition and a no-go theorem.** Confidence is the diagonal term of the inverse-folding log-likelihood and leverage is the mixed second derivative, and we prove leverage is not a function of the bound distribution. A flexible learner over the entire bound distribution recovers only about 37% of it, so about 63% is irreducible from P in both inverse-folding families.
- **A feature-class law.** On natural complexes (SKEMPI) every scalar of the bound distribution (confidence, negentropy, the scalar KL) sits at or below a calibrated placebo floor, while the mixed derivative adds binding information beyond geometry, beyond evolutionary conservation, and beyond the one-pass log-odds, across four architectures.
- **The published deficit is largely a burial confound.** Under a pre-registered matched-pair design the crystal-backbone hotspot deficit attenuates sharply across five architectures and under ProBID-Net’s own released model.
- **Steering a frozen model.** A $+\alpha \cdot L$ tilt on a frozen ProteinMPNN raises binding-favorability, confirmed by four inverse-folding judges, two structure predictors, and a physics energy function (FoldX), on crystal and on predicted backbones.
- **Generality.** The blindness extends from binding to catalytic residues, and the recipe is the classifier-free-guidance direction for any conditional generative model.

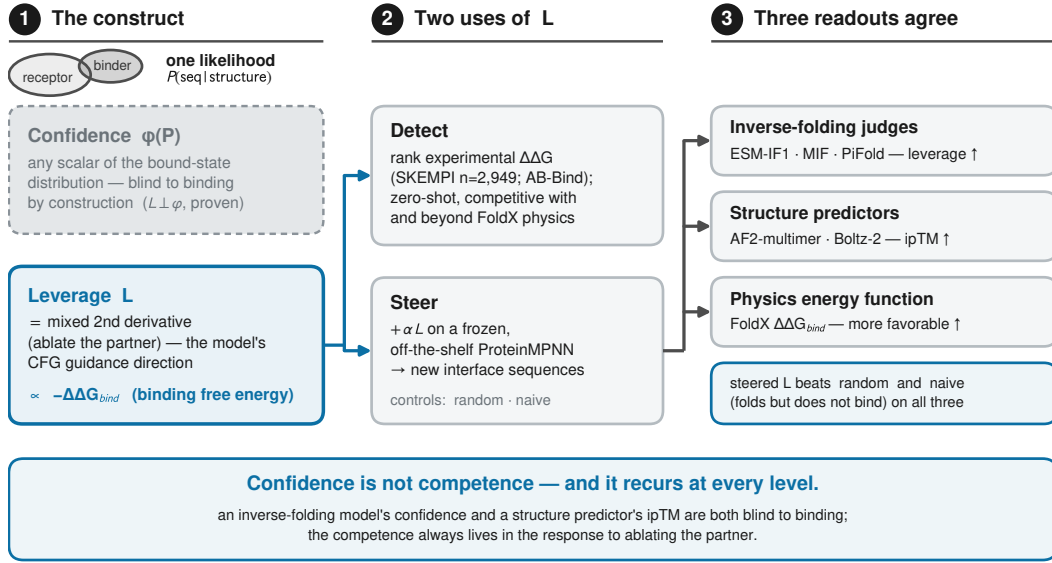


Figure 1: Confidence and competence are two derivatives of one inverse-folding likelihood. Confidence is the diagonal, a scalar of the bound distribution, and it measures fold-stability constraint. Leverage is the mixed second derivative, the response to ablating the partner, and it measures binding. The field has read the diagonal, where binding does not live.

2 SETUP AND PRE-REGISTRATION

Fixtures. Our primary fixture is SKEMPI 2.0 (Jankauskaite et al., 2019), from which we take single-mutation binding data and define a hotspot as an alanine-scan $\Delta\Delta G_{\text{bind}} > 1$ kcal/mol (strict variant > 2 , ProBID-Net’s threshold), with a null set of $|\Delta\Delta G| < 0.25$. Hotspots are rare, 2.4% of interface positions (327/13,401, label entropy 0.115 nats), which sets the scale for the effect sizes below. Because SKEMPI complexes are crystal structures of *natural* complexes, we add two fixtures of different character. Bennett-2023 de-novo designed binders (Bennett et al., 2023) carry site-saturation binding measurements over four targets, a true design-regime test. AB-Bind (Sirin et al., 2016) gives antibody-antigen $\Delta\Delta G$ over 27 analysed complexes, a second SKEMPI-class fixture with distinct biophysics.

The matched-pair protocol, and leakage. The burial analysis rests on a within-complex optimal 1:1 matching of hotspots to null residues on relative complex SASA (± 0.05), secondary-structure class, and neighbour count (± 1). All matching and all cross-validation are *complex-level*: a complex is never split across train and test, so a leak through near-identical interface structures cannot inflate a result. Effects are aggregated by complex-clustered bootstrap, every seed is fixed (20260803) and reported with its replicate count.

Statistical stance. Headline claims are pre-registered, so their tests are confirmatory rather than selected after seeing the data. The placebo floor calibrates the conditional-predictive-impact (CPI) estimator’s false-positive rate directly: a feature must clear the floor, not merely exceed zero. Every interval is a complex-clustered bootstrap. The secondary and cross-architecture analyses are reported without a global family-wise correction and are treated as descriptive support, not independent confirmatory tests.

Models. Five inverse-folding architectures span the design space: ProteinMPNN (Dauparas et al., 2022) (vanilla and soluble), ESM-IF1 (Hsu et al., 2022) (a 142M-parameter GVP-transformer), PiFold (Gao et al., 2023) (a one-shot GNN), MIF (Yang et al., 2023) (masked inverse folding), and ProBID-Net (Chen et al., 2024) (a voxel CNN). A positive control gates every scoring path. These are inverse-folding *sequence* models, the stage our critique targets. Backbone generators such as

RFdiffusion (Watson et al., 2023) are the upstream stage and are not steerable by $+\alpha \cdot L$ nor usable as a leverage judge, so they enter only through the generated-backbone regime we test directly in Section 6.

What the evaluation answers. Four questions organise the empirical sections. **(Q1)** Can any scalar a designer reads off an inverse-folding model (recovery, confidence, entropy, a complex-vs-monomer KL) locate binding hotspots beyond cheap geometry (Section 3)? **(Q2)** If not, where does the model’s binding knowledge live, and is it real beyond geometry, beyond conservation, and beyond the one-pass log-odds, on the natural complexes the field benchmarks on (Section 4)? **(Q3)** Is that knowledge actionable, can it steer a frozen model, and do independent models agree the result binds better (Section 4)? **(Q4)** Does it survive the predicted backbones designers condition on, and is the published recovery deficit a real binding effect or a burial confound (Sections 5 and 6)?

3 CONFIDENCE IS NOT COMPETENCE

The quantity a practitioner is most tempted to trust, the model’s own confidence that the native residue belongs at a position, is useless for locating hotspots. On SKEMPI interface positions, ProteinMPNN’s per-residue confidence attains an AUROC of 0.538 for hotspots, barely above chance, and for a fixed-budget triage (the top-3 interface positions per complex) it captures fewer hotspots than random selection (0.064 vs 0.084, overlapping intervals, so a trend rather than a significant deficit). The point is that confidence buys nothing over random.

One might object that any single scalar can be rescued by combining it with structure. It cannot. We apply a CPI test: cross-fit a model of hotspot-ness on cheap geometry (burial, neighbour count, Δ SASA), then measure how much predictive information confidence adds once its geometry-conditional information is destroyed by permutation within geometry strata. The estimate is 0.000 $[-0.0003, +0.0003]$: confidence is conditionally independent of hotspot-ness given structure. By the same test, Δ SASA adds real information (+0.013) and the KL detector adds a token amount (+0.002).

This is not a quirk of one network. Across all five architectures, interface-hotspot confidence-AUROC lies in 0.50 to 0.54 (ProteinMPNN 0.538, ESM-IF1 0.517, PiFold 0.499, MIF 0.509, ProBiD-Net 0.536), each 0.15 to 0.19 below what trivial burial alone achieves. What predicts hotspots is free geometry: burial reaches 0.689, Δ SASA (the partner-contact area, computable without any network) 0.673, and a cheap-geometry combination 0.734. The learned alternative, a sequence-free KL detector between the complex- and monomer-conditioned distributions, captures only a small signal (CPI +0.002, within-stratum AUROC 0.58, about six times smaller than Δ SASA). None of this is specific to SKEMPI: on AB-Bind the identical pattern holds (confidence-AUROC 0.560 at chance, adding +0.008 over full geometry).

4 THE CONFIDENCE–LEVERAGE DECOMPOSITION: THE MODEL KNOWS BINDING IN THE MIXED DERIVATIVE

The results so far are corrective, and every quantity the field reads off these models (recovery, confidence, the KL) is a scalar summary of the distribution. There is a specific reason they all fail at hotspots, and stating it turns the corrections into a theorem.

4.1 THE DECOMPOSITION AND THE NO-GO

Write the model’s per-position information as two terms of the inverse-folding log-likelihood. **Confidence** is the *diagonal*, a scalar functional of the single bound-conditioned distribution $p(\cdot | X_{\text{complex}})$ (log $p(\text{native})$, negentropy), and estimates fold-stability constraint. **Leverage** is the *mixed second difference*, the per-substitution response to ablating the partner:

$$L_i(a) = [\log p(a | X_{\text{complex}}) - \log p(\text{wt} | X_{\text{complex}})] - [\log p(a | X_{\text{monomer}}) - \log p(\text{wt} | X_{\text{monomer}})],$$

which by the standard binding thermodynamic cycle estimates $-\Delta\Delta G_{\text{bind}}$ up to a single unknown positive scale. This is exactly the classifier-free-guidance direction: a guided logit $\text{logit}(\cdot | X_{\text{complex}}) + \alpha \cdot [\text{logit}(\cdot | X_{\text{complex}}) - \text{logit}(\cdot | X_{\text{monomer}})]$ has bracket equal to L up to a reference

Table 1: The feature-class law on SKEMPI (per interface position, CPI beyond burial, neighbours, and Δ SASA, 13,401 positions). Every scalar of the bound distribution sits at or below the calibrated placebo floor. Only the two-pass mixed derivative clears it.

Feature (all functionals of the same distribution)	CPI beyond geometry
Placebo floor (a deterministic function of geometry)	+0.0007 (false-positive floor)
Confidence (the diagonal)	+0.0002 [−0.0002, +0.0006] (below floor)
Negentropy (one-pass)	+0.0009 [+0.0003, +0.0015] (at floor)
Scalar KL (a contraction of L)	+0.0010 [+0.0003, +0.0017] (at floor)
Leverage L (the mixed derivative)	+ 0.0049 [+0.0034, +0.0065] (about $7\times$ floor)

shift the softmax absorbs. The binding partner is the conditioner and confidence is the conditional marginal, so the recipe for reading an un-trained quantity off a conditional generative model is to ablate the conditioner and take this mixed derivative, not the marginal.

Proposition 1 (Confidence is blind to leverage). Fix a model and an interface position i , with $P_i = p(\cdot \mid X_{\text{complex}})$ and $Q_i = p(\cdot \mid X_{\text{monomer}})$. **(i, cycle)** $L_i(a)$ estimates $-\Delta\Delta G_{\text{bind}}(i, \text{wt} \rightarrow a)$ up to one positive scale. **(ii, non-identifiability)** L is not a function of P : two positions with $P_i = P_j$ have identical confidence $\varphi(P_i) = \varphi(P_j)$ for every functional φ , yet can differ in leverage, so any P -only estimator carries an error floor $\inf_{\varphi} \mathbb{E}[(\Delta\Delta G_{\text{bind}} - \varphi(P))^2] \geq \mathbb{E}[\text{Var}(\Delta\Delta G_{\text{bind}} \mid P)] > 0$. **(iii, readout, not model)** X_{monomer} is X_{complex} with the partner deleted, a deterministic map, so L is computable by a second forward pass yet by (ii) is not a function of P . The binding term is in the weights, reachable only by re-querying on X_{monomer} .

The no-go is not vacuous, and we measure it. A flexible learner (the better of gradient boosting and random forests, out-of-sample under complex-clustered cross-validation) trained on the full 20-vector P recovers only $R^2(L \mid P) = 0.37$ [0.34, 0.40] for ProteinMPNN and 0.36 [0.34, 0.39] for ESM-IF1, so about 63% of the mixed derivative is irreducible from P in both families (about 62% even when wild-type identity is added). A linear readout recovers less than half of that. Against ground truth the floor is starker: regressing experimental $\Delta\Delta G_{\text{bind}}$ directly on P leaves 88% irreducible for ProteinMPNN and 90% for ESM-IF1.

4.2 THE FEATURE-CLASS LAW ON NATURAL COMPLEXES

On SKEMPI, the mixed derivative adds binding information beyond cheap geometry where every scalar summary does not (Table 1, per interface position, CPI over burial, neighbours, and Δ SASA). At the mutation level the effect is large: $\text{Spearman}(L, \Delta\Delta G) = -0.30$ and $\text{CPI}(L \mid \text{geometry}) = +0.059$ [+0.046, +0.073]. The second pass earns its keep against the standard one-pass complex log-odds: controlling for it and geometry, leverage still tracks $\Delta\Delta G$ (partial Spearman -0.147 [−0.190, −0.108]), and the one-pass readout retains signal after controlling for leverage (-0.094), so neither pass subsumes the other. The increment is not conservation in disguise: scoring each position with the field-standard ESM-2 (Lin et al., 2023) masked-marginal estimator (itself the stronger baseline, adding +0.0064 beyond geometry), leverage’s contribution is undiminished, $\text{CPI}(L \mid \text{geometry} + \text{conservation}) = +0.0059$ [+0.0031, +0.0097]. As a zero-shot readout of a model never trained on binding, L adds +0.030 interface AUROC on top of a *supervised* geometry-plus-substitution baseline fit on the labels, which the ground-truth floor predicts: with 88% of experimental $\Delta\Delta G$ irreducible from P , there is little for a supervised model to fit.

All comparators run under identical settings (temperature, decoding-order averaging, and sampling depth held fixed), so the contrasts isolate the feature, not the protocol. The law replicates across four inverse-folding architectures: ESM-IF1 (Spearman -0.26 , CPI +0.035), PiFold (-0.33 , +0.050), and MIF (-0.27 , +0.058), each surviving substitution, confidence, and scalar-KL controls, with confidence position-blind throughout. The law is about a feature class, not a regime: on natural complexes scalar summaries reduce to geometry (confidence exactly, KL nearly), while the mixed derivative does not.

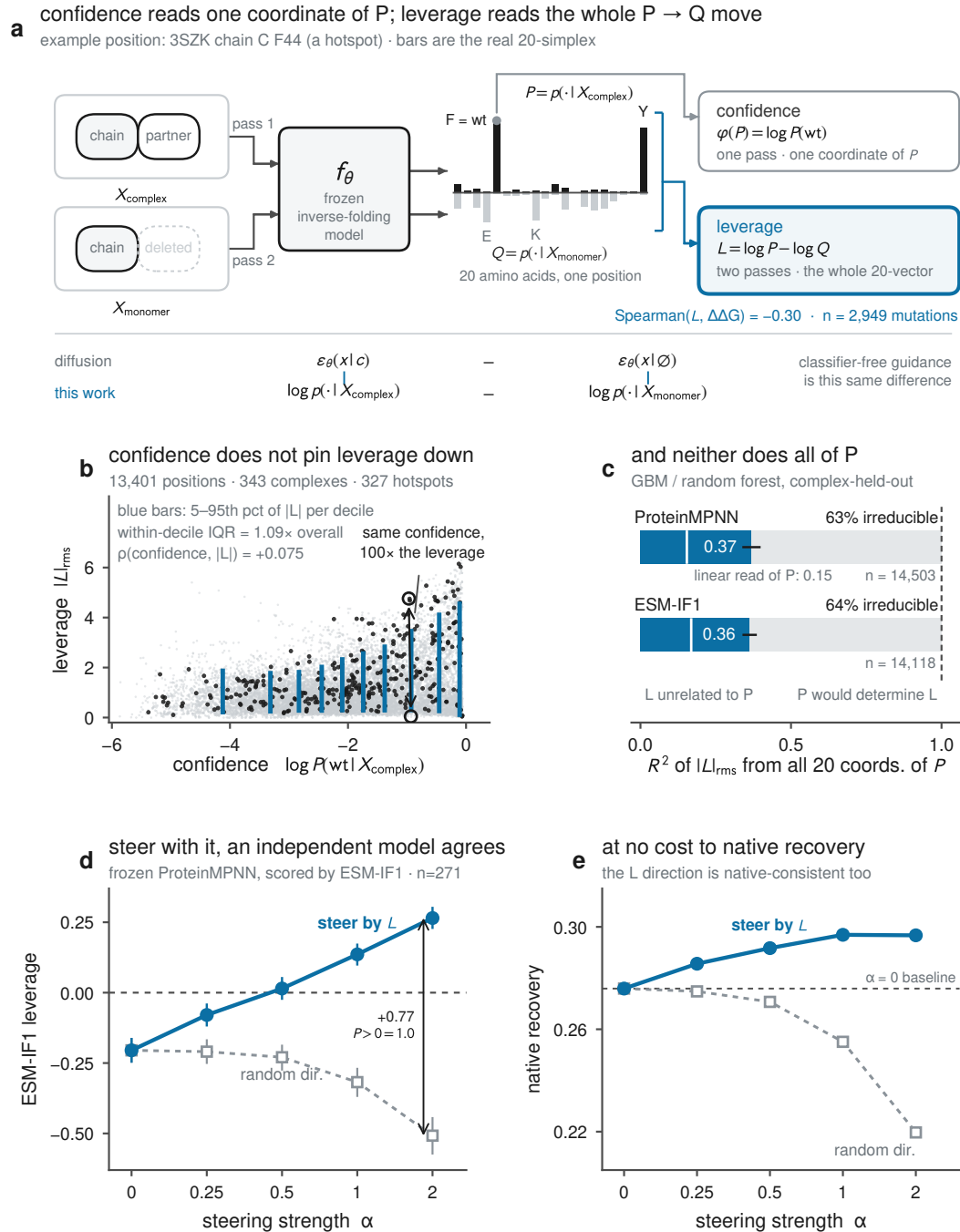


Figure 2: The Confidence-Leverage Decomposition. A flexible learner over the entire bound distribution recovers only about 37 percent of the leverage, so most of the mixed derivative is irreducible from the bound distribution, in both inverse-folding families.

4.3 A DOSE LAW: FRAGILITY TO BACKBONE ERROR

The mixed derivative is read off a backbone, and it does not survive a large perturbation of one. Jittering the crystal backbone to a target interface RMSD and re-scoring (the monomer inheriting the same jitter, so the partner ablation stays clean), $\text{CPI}(L | \text{geometry})$ and $\text{Spearman}(L, \Delta\Delta G)$ hold through sub-Ångström error, then collapse past a model-dependent knee (Table 2). The decisive test

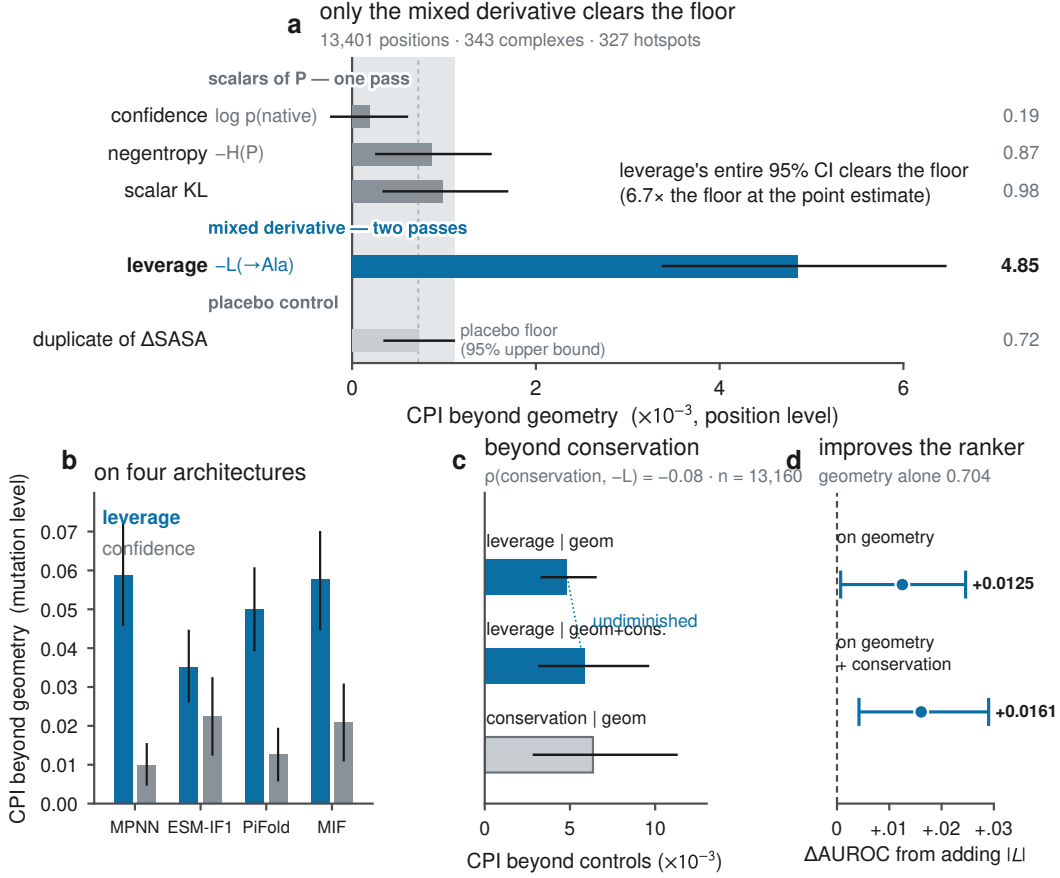


Figure 3: The feature-class law. Every scalar of the bound distribution (confidence, negentropy, the scalar KL) sits at or below the calibrated placebo floor. Only the two-pass mixed derivative clears it.

Table 2: Backbone-error dose law (crystal-jitter ladder). $\dagger$ marks a 95% CI excluding zero (above the $+0.0007$ placebo floor). $n = 2,949$ mutations (ProteinMPNN, matched sample), 2,809 (ESM-IF1, all 285 complexes), seed 20260803.

Jitter σ (Å)	MPNN: CPI beyond geom	MPNN: $\rho(L, \Delta\Delta G)$	ESM-IF1: CPI beyond geom	ESM-IF1: $\rho(L, \Delta\Delta G)$
0.0 (crystal)	$+0.058^\dagger$	-0.30	$+0.036^\dagger$	-0.25
0.25	$+0.059^\dagger$	-0.29	$+0.035^\dagger$	-0.17
0.50	$+0.047^\dagger$	-0.29	$+0.027^\dagger$	-0.17
0.75	$+0.032^\dagger$	-0.19	$+0.011^\dagger$	-0.12
1.0	$+0.002$	-0.08	$+0.018^\dagger$	-0.17
1.5	-0.001	-0.06	$+0.002$	-0.10
2.0	n/a	n/a	$+0.001$	-0.10

is whether real predicted backbones fall on the surviving or the collapsed part, and they fall on the surviving part (Section 6): on OpenFold3 (Ahdriz et al., 2024) and AF2-multimer (Jumper et al., 2021) backbones for 140 shared complexes, $\text{CPI}(L \mid \text{geometry})$ is $+0.039$ [$+0.026, +0.050$] and $+0.032$ [$+0.023, +0.041$], 69 to 84% of the matched crystal value. Because the collapse is driven by the backbone the derivative is read from, the same fragility is a prediction every method built on this mixed derivative inherits on predicted backbones, and the threshold is model-dependent (about 1.0 Å for ProteinMPNN, about 1.5 Å for ESM-IF1), not a universal cliff.

The knowledge also reaches the second mixed derivative. The partner-ablated pairwise coupling tracks measured binding epistasis with the cycle-predicted sign, $\text{Spearman}(C, g) = -0.14$

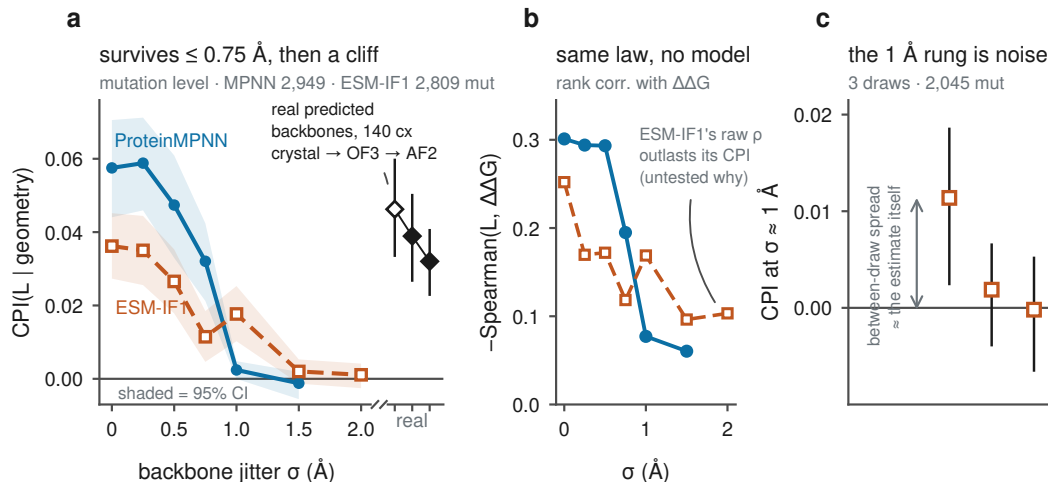


Figure 4: The backbone-error dose law. The mixed derivative survives sub-Ångström reconstruction error and collapses past a model-dependent knee. Real predicted backbones fall on the surviving part.

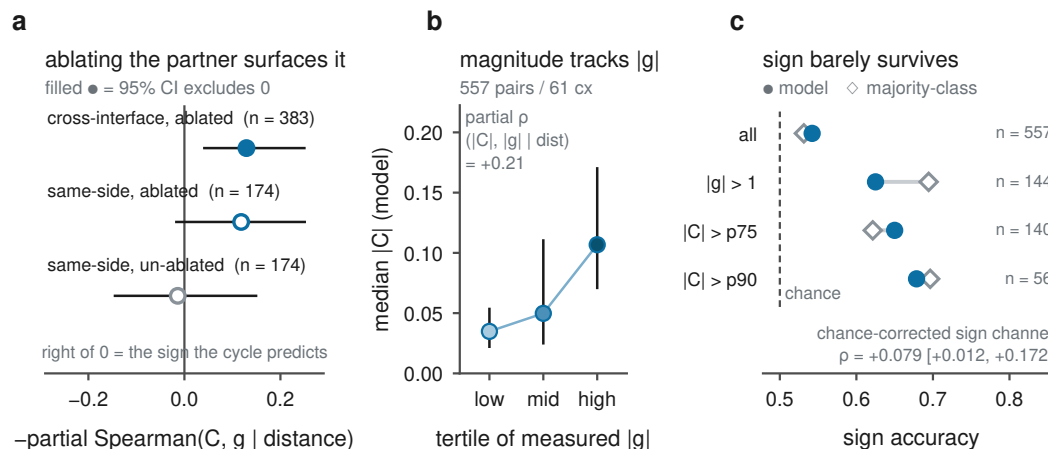


Figure 5: The second mixed derivative. The partner-ablated pairwise coupling tracks measured binding epistasis with the cycle-predicted sign, surviving control for inter-residue distance.

$[-0.23, -0.07]$, surviving control for inter-residue distance (partial -0.12). The effect is modest and is a signal about coupling magnitude more than sign, and partner ablation is what surfaces it: the un-ablated coupling is null. On de-novo designs (Bennett-2023, site-saturation labels) even the scalar complex-conditioned distribution ranks substitutions by binding (interface AUROC 0.615, above every substitution baseline), answers the stability question better than the binding question (0.721 vs 0.615, the control a fold-trained model should pass), and adds +0.018 beyond an all-atom rotamer-repacked occlusion baseline, reproduced by a non-parent scorer.

4.4 STEERING A FROZEN MODEL

If L is the guidance direction, the direct test is to guide with it. Biasing a frozen, off-the-shelf ProteinMPNN's interface logits by $+\alpha \cdot L$ and sampling ($K=64$, 271 complexes, pre-registered grid) raises the mean binding-leverage a *different* model (ESM-IF1) assigns the sampled residues, monotonically, from -0.20 at $\alpha = 0$ to $+0.27$ at $\alpha = 2$. A matched-magnitude random direction does the opposite, so the paired L - random gap reaches $+0.77$ ($P(> 0) = 1.0$ at every $\alpha > 0$): it

Table 3: Steering benefit across independent readouts (paired L – random, complex-clustered 95% CI). ipTM rows are within-folder paired effects (AF2 and Boltz-2 are differently calibrated, each read in its own scale). Judge rows are on the leverage scale. Self-judged cells (steer-X, judge-X) are excluded as trivially circular.

Steered model	Independent readout	Paired L – random
ProteinMPNN	AF2-multimer ipTM ($n = 120$)	+0.235 [+0.199, +0.272]
ProteinMPNN	Boltz-2 ipTM ($n = 60$)	+0.139 [+0.101, +0.177]
ProteinMPNN	ESM-IF1 leverage (judge)	+0.77 [+0.72, +0.82]
ProteinMPNN	MIF leverage (judge)	+0.71 [+0.66, +0.76]
ProteinMPNN	PiFold leverage (judge)	+0.89 [+0.84, +0.95]
ESM-IF1 (reverse)	AF2-multimer ipTM ($n = 60$)	+0.158 [+0.102, +0.215]
ESM-IF1 (reverse)	ProteinMPNN leverage (judge)	+0.43 [+0.37, +0.49]
ESM-IF1 (reverse)	MIF leverage (judge)	+0.57 [+0.48, +0.66]
ESM-IF1 (reverse)	PiFold leverage (judge)	+0.73 [+0.62, +0.84]

is the direction, not the perturbation. Native interface recovery does not fall but slightly rises (0.276 to 0.297). Scoring with a different model is what makes this non-trivial, and the steered sequences transfer to an independent structure predictor: folded with AF2-multimer (120 complexes), the L -steered interfaces beat the random control on ipTM by +0.235 [+0.199, +0.272], interface-localized (global pTM rises about $2.9\times$ less). The effect is robust across a matrix of independent readouts (Table 3): a second structure predictor (Boltz-2 (Passaro et al., 2025)), a second steered model (ESM-IF1, judged by a third), and four anti-circular judges, with the trivially-circular self-judged cells excluded.

A specificity control bounds the interpretation. A *naive* tilt, the model’s own log-odds direction rescaled to L ’s magnitude (be more confident, not bind better), recovers most of the ipTM gain and slightly exceeds L there, because ipTM is itself a structure-level confidence and rewards a foldable, confident interface. This is the same confidence-is-not-competence split one level up. The binding-specific increment survives where the mechanism predicts, at the inverse-folding-judge level, where L beats the confidence tilt (ESM-IF1 +0.159 [+0.110, +0.207], MIF +0.242 [+0.191, +0.292]). The confidence tilt is a strong but binding-*correlated* baseline (it carries 64 to 77% of L – random at the judge level), not binding-*specific*: a confidently-packed interface is a prerequisite for binding, so the confidence direction rides along with binding without targeting it. In the classifier-free-guidance framing, naive is the unguided marginal and L is the guidance term, so “does L beat naive” is exactly “does the guidance add over the marginal,” which is why naive, not random, is the control the framing demands.

4.5 PHYSICS ADJUDICATION (FOLDX)

To close the gap between inverse-folding judges and a real binding term, we ran FoldX (Guerois et al., 2002; Schymkowitz et al., 2005; Delgado et al., 2019), a hand-fit physics energy function with an explicit binding term, from a modality outside inverse folding (pre-registered, falsifier: L – naive favorability ≤ 0). On the 60-complex steering set, L beats the confidence tilt on physics: a designer keeping the best of a few samples gets L – naive = +1.86 kcal/mol [+0.72, +3.02] (best-of-k, the design-relevant yield), and the pre-registered mean-over-k primary is +1.24 [+0.14, +2.33], both clearing zero (Table 4). Because FoldX is physics, not an inverse-folding proxy, it sides with the judges ($L > \text{naive}$) against ipTM (naive $\gtrsim L$), so the binding-specific advantage of L is real, and the ipTM disagreement was the foldability effect the frustration thesis predicts. As a detector, FoldX (fit to $\Delta\Delta G$) ranks better than the zero-shot L ($|0.43|$ vs $|0.30|$), and we do not claim otherwise. The narrower and more surprising claim is that L carries binding rank-signal beyond the fitted energy function: partial Spearman($L, \Delta\Delta G \mid \text{FoldX}$) = -0.17 [$-0.23, -0.12$]. The direction replicates on a second fixture (AB-Bind), same-signed but underpowered at 22 crystal complexes.

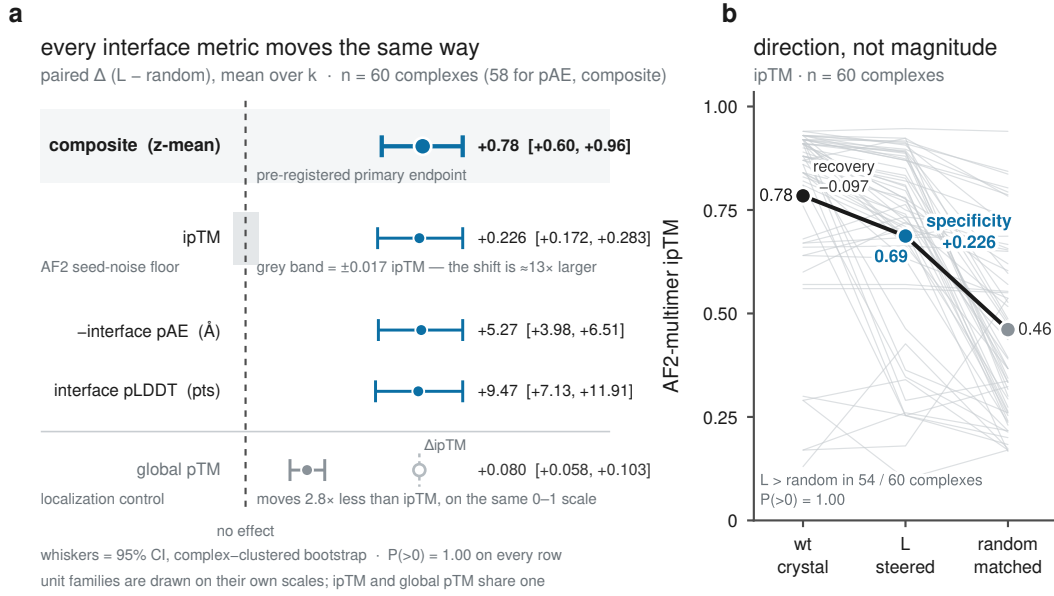


Figure 6: Steering a frozen ProteinMPNN by the mixed derivative raises AF2-multimer interface ipTM against a matched-magnitude random control, and the gain is interface-localized (global pTM rises far less).

Table 4: FoldX adjudication. Left: detection ($n = 2,948$ mutants, 284 complexes), Spearman vs experimental $\Delta\Delta G$. The two opposite signs both mean “ranks binding correctly”: experimental $\Delta\Delta G$ and FoldX score cost, L scores favorability. Right: steering specificity (best-of- k , paired 95% CI).

Detection	Spearman vs exp $\Delta\Delta G$
FoldX $\Delta\Delta G_{\text{bind}}$ (physics, fit)	+0.434 [+0.38, +0.49]
L (zero-shot inverse-folding)	−0.301 [−0.35, −0.24]
L FoldX (partial, beyond physics)	−0.17 [−0.23, −0.12]
Steering specificity (kcal/mol)	Δ , 95% CI, $P(> 0)$
L – naive (decisive)	+1.86 [+0.72, +3.02], 0.999
L – random	+6.27 [+5.03, +7.49], 1.000
naive – random	+4.79 [+3.46, +6.11], 1.000

4.6 PREDICTED-BACKBONE STEERING (THE DESIGN REGIME)

Every steering result so far tilts the model on a crystal backbone, but the staged-design regime has only a predicted one. So we recomputed the leverage and applied the tilt on the predicted backbone (OpenFold3 and AF2-multimer, 106 complexes), judging anti-circularly by an independent model’s leverage also read off the predicted structure (pre-registered, falsifier: judge-level L – random ≤ 0). The falsifier does not fire and the effect is barely attenuated: judge-level L – random is +0.70 to +0.76 across both predicted backbones and both judges, essentially equal to the matched crystal control, and L – naive stays positive (+0.17 to +0.22), so specificity is preserved. An independent structure predictor confirms it: on 93 complexes, interface ipTM L – random is +0.18 [+0.14, +0.22] (Figure 8). This is one dose law spanning both uses of the direction: the design regime that preserves the readout preserves the knob.

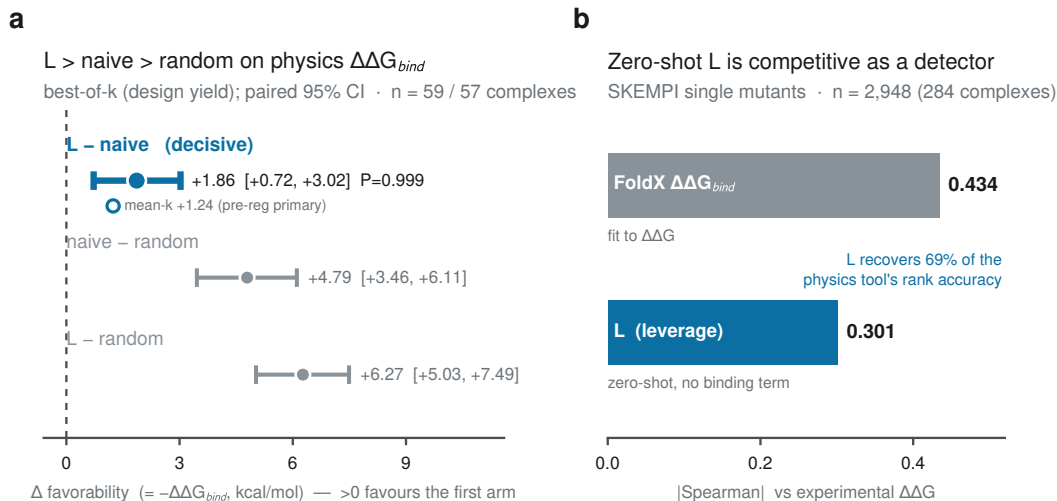


Figure 7: FoldX adjudication. As a detector the fitted physics ranks better than the zero-shot leverage, yet the leverage carries binding rank-signal beyond it (partial correlation). As a steering readout the mixed derivative beats the confidence tilt, so the advantage is binding-specific, not foldability.

5 ON CRYSTAL BACKBONES, THE HOTSPOT GAP IS A BURIAL ARTIFACT

We now return to the published deficit and show, on crystal backbones, that it is a burial confound. The confound is visible directly: as hotspot strength rises, both recovery and burial rise in lockstep (recovery 0.347 to 0.529, relative SASA 0.218 to 0.080). Under the matched-pair design the deficit attenuates to statistical indistinguishability: the matched estimate is -0.042 $[-0.222, +0.129]$ and a higher-powered regression estimator is $+0.059$ $[-0.051, +0.167]$. The matching does not control partner-contact area, so we check the residual directly: adjusting for the Δ SASA imbalance moves the deficit essentially nowhere (-0.042 to -0.028). No architecture retains a hotspot deficit under matching. Most attenuate to zero, and two (MIF, PiFold) flip to a hotspot advantage on the under-powered strict tier. The strongest form of the test uses ProBID-Net’s own released voxel-CNN: its port is faithful (interface recovery 0.472), its published deficit reproduces only in comprehensively-scanned complexes, and it dissolves under confound-matching (matching residue type turns it positive, burial gives -0.038 , hydrophobicity -0.051 , every interval spanning zero). ProBID-Net’s deficit is a residue-composition and burial confound, and we offer the matched-pair design as a reusable diagnostic protocol.

6 THE TAX LIVES IN THE CONDITIONING SET

If the deficit were a benchmark artifact it would vanish once burial is controlled. It does not. It reappears on the predicted backbones designers condition on. On backbones from two architecturally-independent predictors (OpenFold3 and AF2-multimer) a burial-matched deficit is present (-0.191 and -0.233 , the crystal deficit is about zero). The claim rests not on either marginal number but on their agreement: the two predictors’ per-complex deficits correlate at $\rho = 0.565$ $[0.40, 0.71]$, the same three complexes carry the effect under both, and the agreement survives residualising on interface burial (partial $+0.529$). The effect tracks how a backbone is non-native, not how far: on noised crystals at matched RMSD the deficit is absent, so it is a property of independent reconstruction. And the binding signal a designer would read off these backbones survives them: recomputed on the predicted structures, $CPI(L \mid \text{geometry})$ is $+0.039$ and $+0.032$ (pooled $+0.036$, $P(> 0) = 1.0$), a 69 to 84% retention of the crystal value, with the positive control reproducing the committed leverage to 10^{-5} . The confidence readout degrades into the deficit while the mixed derivative survives, exactly where the decomposition says they must diverge.

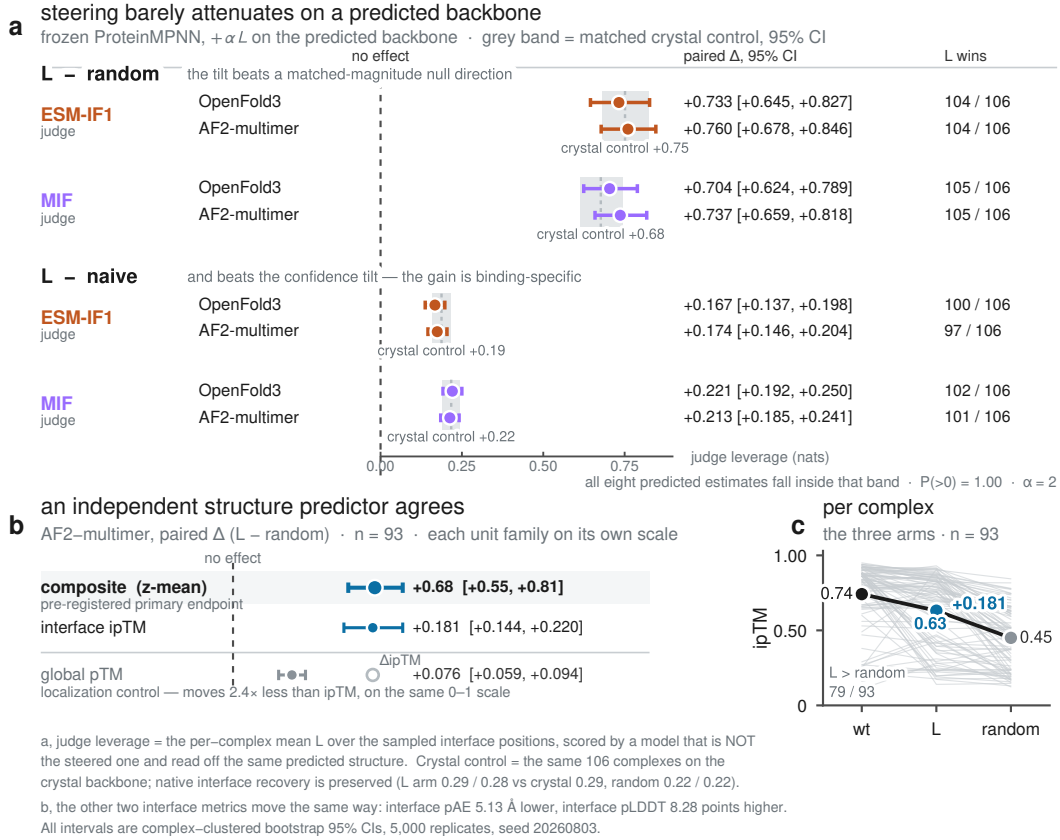


Figure 8: Steering survives the swap to a predicted backbone. **a**, judge-level paired L – random and L – naive on OpenFold3 and AF2-multimer backbones under two anti-circular judges. The grey band is the matched crystal control’s 95% CI on the same 106 complexes, and all eight predicted estimates fall inside it. **b**, an instrument outside the inverse-folding family agrees, with global pTM as the localization control. **c**, the same ipTM effect per complex.

7 RULING OUT COMPETING MECHANISMS

Two alternative explanations are separately measured and refuted. A *sample-budget* effect (at low sampling temperature a multi-residue hotspot constellation has astronomically small joint probability) is identical at burial-matched control constellations (median difference 0.000, 95% CI $[-1.67, +2.09]$ spanning zero): it is a generic property of low-temperature sampling, not a hotspot tax, and no oversampling recovers it. A *commitment-ordering* effect (the autoregressive schedule paints hotspots into a corner) is inert on ProteinMPNN (oracle-order difference-in-differences -0.002), and on MultiFlow the unmasking-order knob has only a marginal effect (order-span 0.012, far below the 0.065 sample SD across seeds and reference structures). What remains is the conditioning-set signal of Section 6.

8 RELATED WORK AND POSITIONING

The nearest prior methods use this operator, so we are exact about the boundary. The leverage operator L is BA-Cycle (Jiao et al., 2025), whose bound-versus-unbound double-difference rearranges to our mixed second difference, and we credit it for the score. RedNet independently operationalises the same contrast as a design-time decoder, verified in its released code, and we credit it too. What is not theirs and is ours is a different object. (i) The decomposition identifies their score as the mixed derivative and confidence as the diagonal, with the constructive proof that confidence, and every scalar of P , is blind to it, a question neither poses. (ii) We give the first beyond-geometry and

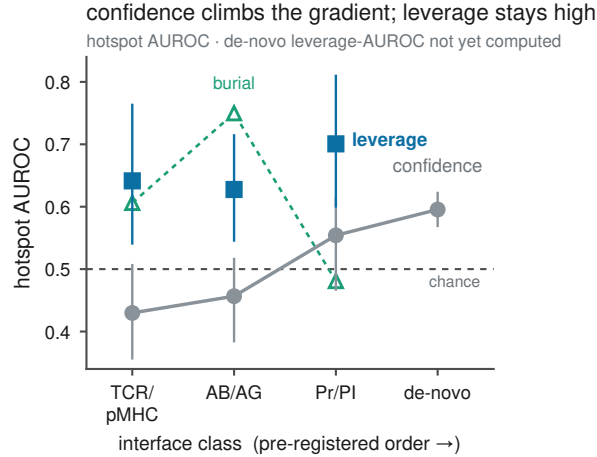


Figure 9: Confidence-AUROC climbs a fold-coupling gradient across interface classes and only clears chance on de-novo binders, while the mixed derivative clears chance in every class. The scalar is regime-dependent and the derivative is not.

beyond-conservation control for an inverse-folding binding signal, built on CPI (Watson & Wright, 2021) with a conditional permutation test (Berrett et al., 2020), which BA-Cycle and RedNet run in no form. (iii) We state the feature-class law. (iv) We show frozen-model steering, where RedNet retrains a decoder and the direction is already a drop-in tilt on a frozen model with native recovery preserved. (v) We validate anti-circularly across four judges, two structure predictors, and a physics energy function, with a confidence-tilt specificity control no design method has run. A frozen-vs-retrained head-to-head is not runnable (RedNet needs non-public packages and access-restricted weights) and would be muddy, since the direction is shared, so the sharp comparisons are the ones we report: L vs random and L vs the confidence tilt.

These add up to a beyond-X ladder (Figure 12): the mixed derivative carries binding signal that survives, in turn, geometry, substitution similarity, the one-pass log-odds, and a fitted physics energy function, while every scalar of P stays at the placebo floor at every rung. Read with the fractal motif of Section 1, these are two axes of one dissociation: a scalar of anything is blind to binding-specificity, across controls and across levels of the modeling stack, and the mixed derivative carries it. The sequence-free KL detector is a learned frustratometer, a concept we credit to the classical frustration literature (Ferreiro et al., 2007; Freiburger et al., 2019), and our narrow point is only that this one scalar equals ΔSASA , which is why it does not beat physics, whereas the mixed derivative does. ProBID-Net reports the phenomenon as a recovery deficit, and we correct the attribution: it is neither dynamics nor decoding but conditioning, and a burial confound on the crystal benchmark. The most telling piece of practice is BindCraft (Pacesa et al., 2025), whose one-shot pipeline hard-codes an interface freeze that forbids inverse folding at the interface, the field’s implicit admission of our thesis. Ranking by confidence is at chance there (which justifies the freeze), but adding the mixed derivative to the standard geometric feature set raises hotspot AUROC over geometry alone (0.704) by a paired $+0.0125$ [$+0.0007$, $+0.0246$], so the corrected rule is to rank by geometry plus the mixed derivative, not by confidence. Independent corroboration comes from Janusz et al. (2026), who report “a very weak effect of antigen on the predictions” in antigen-aware antibody inverse folding, the same partner-insensitivity we quantify.

9 LIMITATIONS

We evaluate on three fixtures (SKEMPI, Bennett de-novo, AB-Bind), none a full generate-design-wet-lab loop, and we are explicit about which half rests on what. The *detection* claim is experiment-anchored: L correlates with experimental $\Delta\Delta G$ (Spearman -0.30) and adds beyond a fitted physics function (partial -0.17), both against real measurements. The *steering* claim’s binding-specificity is anchored, out-of-family, by one approximate-physics readout (FoldX, mean-over-k L – naive

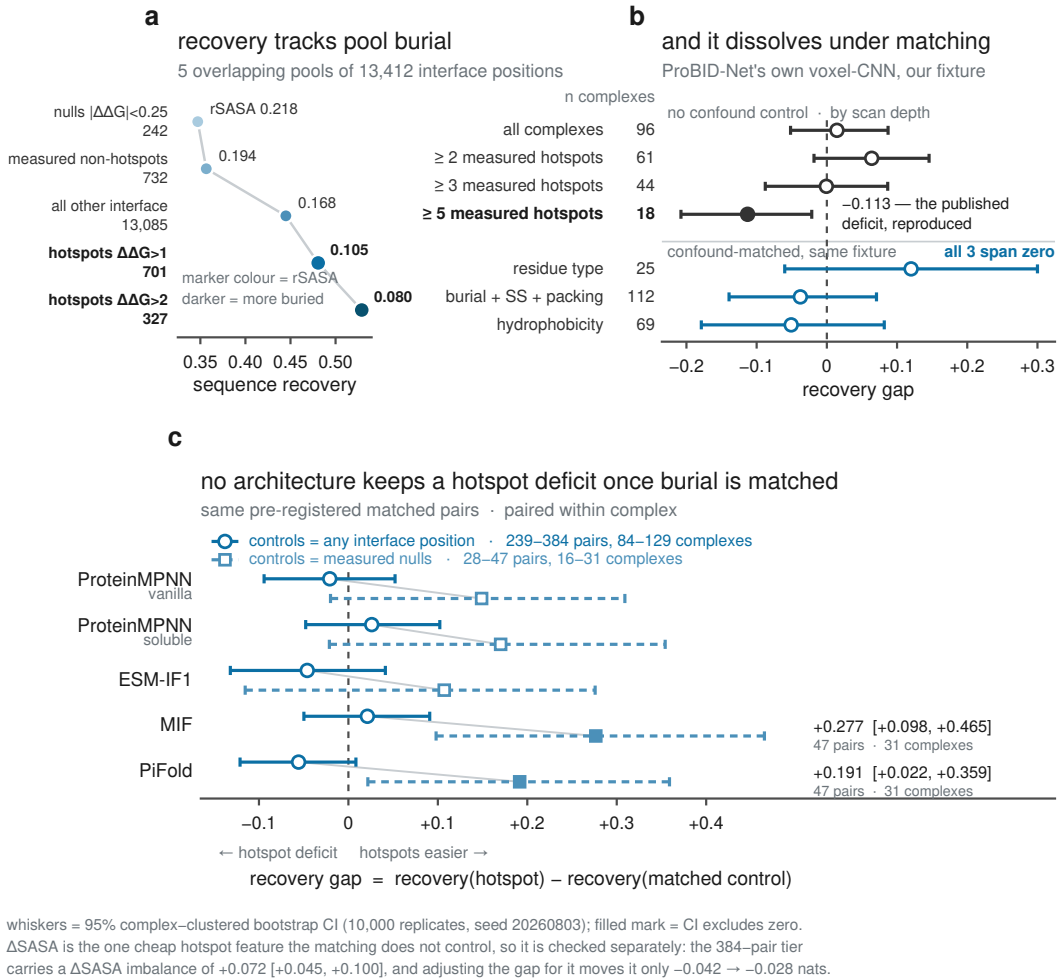


Figure 10: On crystal backbones the hotspot recovery deficit tracks burial. Under the pre-registered matched-pair design it attenuates to statistical indistinguishability, so the published deficit is largely a burial confound.

+1.24, lower CI +0.14), so we scope it as physics-supported pending a wet-lab test rather than experimentally proven, and do not overclaim it.

The primary claims rest on one natural fixture. This is structural to the field’s data, not a gap in our search: SKEMPI 2.0’s curators already absorbed the non-redundant content of the other curated affinity databases, and the subsequent resources are aggregators built around it. We mitigate with two further fixtures, four architectures, a catalytic-site replication, and the cleanest non-overlapping alternative, ATLAS (Borrmann et al., 2017) (TCR-pMHC), where the direction replicates (Spearman(L , $\Delta\Delta G$) = -0.22/ - 0.32 across the two models, bracketing SKEMPI’s -0.30). The direction thus generalizes across three interface classes (protein-protein -0.30, antibody-antigen -0.18, TCR-pMHC -0.22/ - 0.32), though only SKEMPI is powered for the geometry-controlled add-on. Tellingly, ATLAS is the lowest-confidence interface class, so it tests the mixed derivative exactly where the scalar readout is blindest, and the direction still points the right way. The blindness itself extends beyond binding, to catalytic residues (M-CSA (Ribeiro et al., 2018)): inverse-folding confidence is blind (within-amino-acid-type AUROC 0.43 to 0.52) while a sequence language model’s conservation predicts them (0.77), a dissociation of +0.288 that survives composition, burial, and chain-truncation controls.

Why an in-silico result is informative now. The detection half is validated against real $\Delta\Delta G$ measurements on three fixtures, so it is not in-silico in the sense that matters. The steering half is

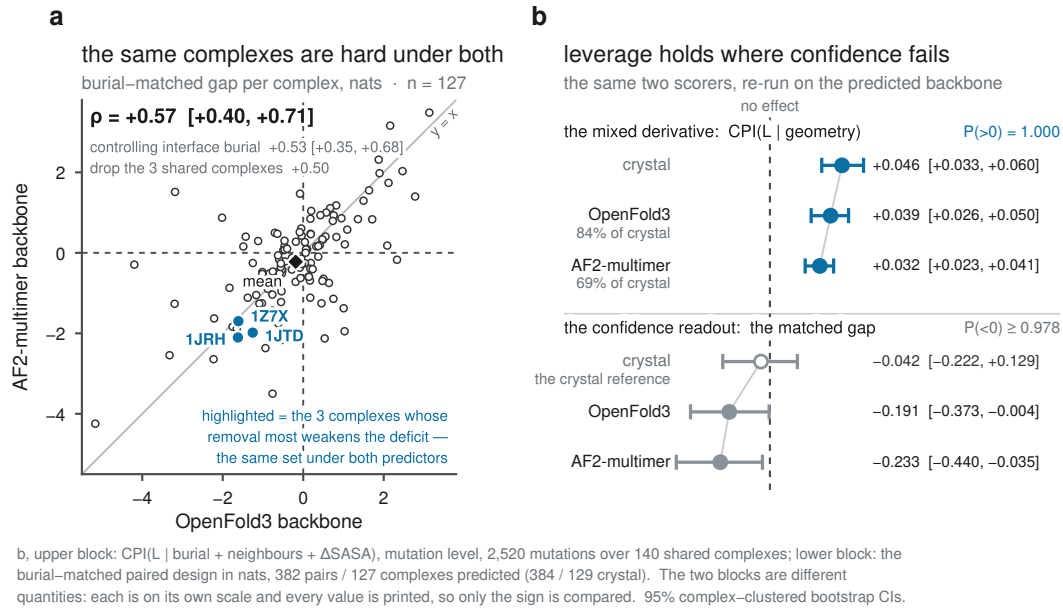


Figure 11: On predicted backbones a burial-matched deficit reappears, and the two architecturally-independent predictors agree per complex, so the residual tax lives in the conditioning set, not in the benchmark.

L clears every control anyone has proposed

one matched metric · SKEMPI interface mutants, n = 2,948–2,949 (284–285 complexes)

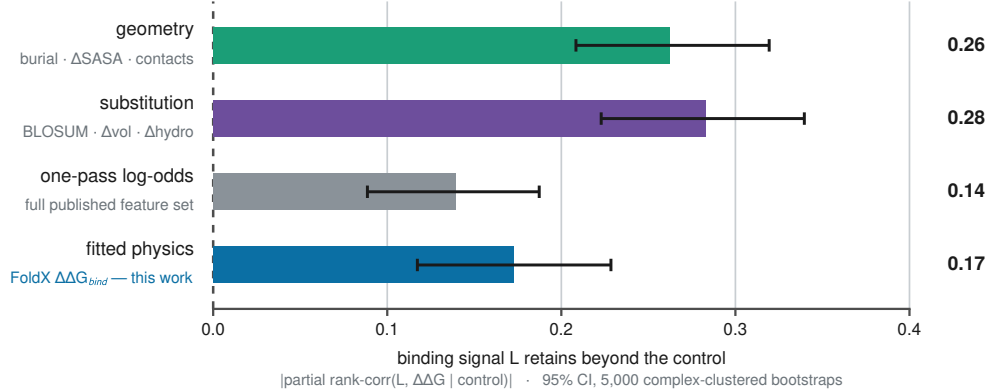


Figure 12: The beyond-X ladder in one matched metric (partial rank-correlation of L with experimental $\Delta\Delta G$, controlling for each competing feature class). L clears geometry, substitution similarity, the one-pass log-odds, and a fitted physics energy function. Every scalar of P stays at the floor.

validated by three instruments that do not share a failure mode (a second sequence model, two structure predictors, and a physics energy function with an explicit binding term), which is the standard by which comparable design methods are currently judged. A wet-lab loop is the right next step and is the one claim we scope as pending, not the foundation the paper stands on.

Caveats specific to the decomposition. Orthogonal is not independent: confidence cannot express leverage, but the two are weakly-to-moderately correlated and the correlation is model-dependent (Spearman(confidence, $|L|$) = +0.075 for ProteinMPNN, +0.30 for ESM-IF1), so we claim blindness by construction, not statistical independence. L estimates $-\Delta\Delta G_{bind}$ only up to a temperature, and all headline readouts are rank-based, but a single 2-parameter out-of-fold affine reads one unit

of L as about 0.42 kcal/mol (OOF RMSE 1.82, next to fitted FoldX's 1.79), a good ranker and a rough absolute predictor, kept out of every headline. The decomposition replicates under a second architecture (ESM-IF1) on the 337 complexes both models cover.

REPRODUCIBILITY STATEMENT

Every numerical result traces to a committed script and CSV, each carrying its exact command and a fixed seed (20260803). A fresh pointer-only clone reproduces every headline number and figure from the committed CSVs alone, with no GPU, no Git-LFS pull, and no cluster access, which we verified directly. The central result (the Confidence-Leverage Decomposition) was additionally subjected to an independent adversarial audit of its statistical machinery: the conditional-independence test was verified not to leak, with pure-noise features giving CPI about zero, a null floor about 16 times below the reported effect.

LLM USAGE DISCLOSURE

Per the ICLR LLM-usage policy, we disclose that large language models were used as an assistant for code generation, data analysis, prose drafting and editing, and research ideation. All experiments, numerical results, and citations were verified and validated by the authors, who are solely responsible for the contents of this submission. No result or citation is included that does not resolve to committed code and data or to a source we fetched and checked.

REFERENCES

- Gustaf Ahdriz et al. Openfold: retraining alphafold2 yields new insights into its learning mechanisms and capacity for generalization. *Nature Methods*, 21(8):1514–1524, 2024. doi: 10.1038/s41592-024-02272-z.
- Nathaniel R. Bennett et al. Improving de novo protein binder design with deep learning. *Nature Communications*, 14(1):2625, 2023. doi: 10.1038/s41467-023-38328-5.
- Thomas B. Berrett, Yi Wang, Rina Foygel Barber, and Richard J. Samworth. The conditional permutation test for independence while controlling for confounders. *Journal of the Royal Statistical Society: Series B*, 82(1):175–197, 2020. doi: 10.1111/rssb.12340.
- Tyler Borrmann et al. Atlas: A database linking binding affinities with structures for wild-type and mutant tcr-pmhc complexes. *Proteins: Structure, Function, and Bioinformatics*, 85(5):908–916, 2017. doi: 10.1002/prot.25260.
- Zhihang Chen, Menglin Ji, Jie Qian, Zhe Zhang, Xiangying Zhang, Haotian Gao, Haojie Wang, Renxiao Wang, and Yifei Qi. Probid-net: a deep learning model for protein-protein binding interface design. *Chemical Science*, 15(47):19977–19990, 2024. doi: 10.1039/d4sc02233e.
- Justas Dauparas et al. Robust deep learning-based protein sequence design using proteinmpnn. *Science*, 378(6615):49–56, 2022. doi: 10.1126/science.add2187.
- Javier Delgado, Leandro G. Radusky, Damiano Cianferoni, and Luis Serrano. FoldX 5.0: working with RNA, small molecules and a new graphical interface. *Bioinformatics*, 35(20):4168–4170, 2019. doi: 10.1093/bioinformatics/btz184.
- Diego U. Ferreira, Joseph A. Hegler, Elizabeth A. Komives, and Peter G. Wolynes. Localizing frustration in native proteins and protein assemblies. *Proceedings of the National Academy of Sciences*, 104(50):19819–19824, 2007. doi: 10.1073/pnas.0709915104.
- Maria I. Freiberger, Ariel B. Guzovsky, Peter G. Wolynes, R. Gonzalo Parra, and Diego U. Ferreira. Local frustration around enzyme active sites. *Proceedings of the National Academy of Sciences*, 116(10):4037–4043, 2019. doi: 10.1073/pnas.1819859116.
- Zhangyang Gao, Cheng Tan, and Stan Z. Li. Pifold: Toward effective and efficient protein inverse folding. In *International Conference on Learning Representations (ICLR)*, 2023.

- Raphael Guerois, Jens Erik Nielsen, and Luis Serrano. Predicting changes in the stability of proteins and protein complexes: a study of more than 1000 mutations. *Journal of Molecular Biology*, 320(2):369–387, 2002. doi: 10.1016/S0022-2836(02)00442-4.
- Chloe Hsu, Robert Verkuil, Jason Liu, Zeming Lin, Brian Hie, Tom Sercu, Adam Lerer, and Alexander Rives. Learning inverse folding from millions of predicted structures. In *International Conference on Machine Learning (ICML)*, 2022.
- Justina Jankauskaite, Brian Jimenez-Garcia, Justas Dapkunas, Juan Fernandez-Recio, and Iain H. Moal. Skempi 2.0: an updated benchmark of changes in protein-protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics*, 35(3):462–469, 2019. doi: 10.1093/bioinformatics/bty635.
- Bartosz Janusz, Dawid Chomicz, Sonia Wrobel, Pawel Dudzic, Adithya Polasa, Kyle Martin, Steven Darnell, Stephen R. Comeau, and Konrad Krawczyk. Benchmarking antigen-aware inverse folding methods for antibody design. *Bioinformatics Advances*, 6(1):vbag081, 2026. doi: 10.1093/bioadv/vbag081.
- Xinyou Jiao, Aoxue Mao, Wengong Jin, Xuezhi Yang, Kangfei Chen, and Yang Shen. Boltzmann-aligned inverse folding model as a predictor of mutational effects on protein-protein interactions. In *International Conference on Learning Representations (ICLR)*, 2025.
- John Jumper et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.
- Zeming Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.
- Martin Pacesa et al. One-shot design of functional protein binders with bindcraft. *Nature*, 646(8084):483–492, 2025. doi: 10.1038/s41586-025-09429-6.
- Saro Passaro, Gabriele Corso, Jeremy Wohlwend, et al. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707.
- Antonio J. M. Ribeiro et al. Mechanism and catalytic site atlas (m-csa): a database of enzyme reaction mechanisms and active sites. *Nucleic Acids Research*, 46(D1):D618–D623, 2018. doi: 10.1093/nar/gkx1012.
- Joost Schymkowitz, Jesper Borg, Francois Stricher, Robby Nys, Frederic Rousseau, and Luis Serrano. The FoldX web server: an online force field. *Nucleic Acids Research*, 33(suppl_2):W382–W388, 2005. doi: 10.1093/nar/gki387.
- Sarah Sirin, James R. Apgar, Eric M. Bennett, and Amy E. Keating. Ab-bind: Antibody binding mutational database for computational affinity predictions. *Protein Science*, 25(2):393–409, 2016. doi: 10.1002/pro.2829.
- David S. Watson and Marvin N. Wright. Testing conditional independence in supervised learning algorithms. *Machine Learning*, 110(8):2107–2129, 2021. doi: 10.1007/s10994-021-06030-6.
- Joseph L. Watson et al. De novo design of protein structure and function with rfdiffusion. *Nature*, 620(7976):1089–1100, 2023. doi: 10.1038/s41586-023-06415-8.
- Xie et al. Redesign selective protein binders using contrastive decoding. *bioRxiv*, 2026. doi: 10.64898/2026.05.09.722041.
- Kevin K. Yang, Niccolò Zanichelli, and Hugh Yeh. Masked inverse folding with sequence transfer for protein representation learning. *Protein Engineering, Design and Selection*, 36:gza015, 2023. doi: 10.1093/protein/gza015.

Table 5: Pre-registered false-positive modes and their controls.

Mode	Control	Result
Burial (buried positions are where inverse folding is most confident, so it hides the effect)	within-complex matched pairs (rSASA, secondary structure, neighbour count) plus CPI over geometry at every step	deficit attenuates across 5 architectures plus ProBID-Net, and leverage CPI +0.0049 survives full geometry
Native amino-acid identity	per-type breakdown, alanine-only subset, substitution-similarity partialled out	Spearman negative in 18/19 wt-types and -0.25 on Ala-only
PDB training leakage	none needed: leakage makes a positive conservative	every positive is a lower bound
Assay heterogeneity	strict (> 2) and loose (> 1) hotspot thresholds, both reported	conclusions hold under both (partial: no temperature/pH stratification, disclosed)
Positional independence	measure the second mixed derivative directly	couplings track epistasis, Spearman -0.14
Decoding-order variance	sequence-free marginals, else symmetrise over ≥ 8 orders	disjoint-order subsets agree +0.61, oracle order inert (-0.002)

A PRE-REGISTERED FALSE-POSITIVE MODES AND THEIR CONTROLS

Six ways to get a false positive were named in the pre-registration before any number was computed (Table 5). One (assay heterogeneity) is controlled only in part, and we say so. The pattern worth noting is that the two largest confounds, burial and training leakage, both run against the hypothesis, so the burden is to show the effect despite them, which is why the matched-pair design, not the raw deficit, is the experiment.